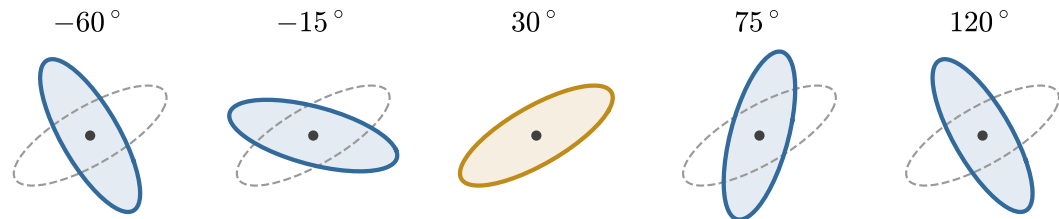
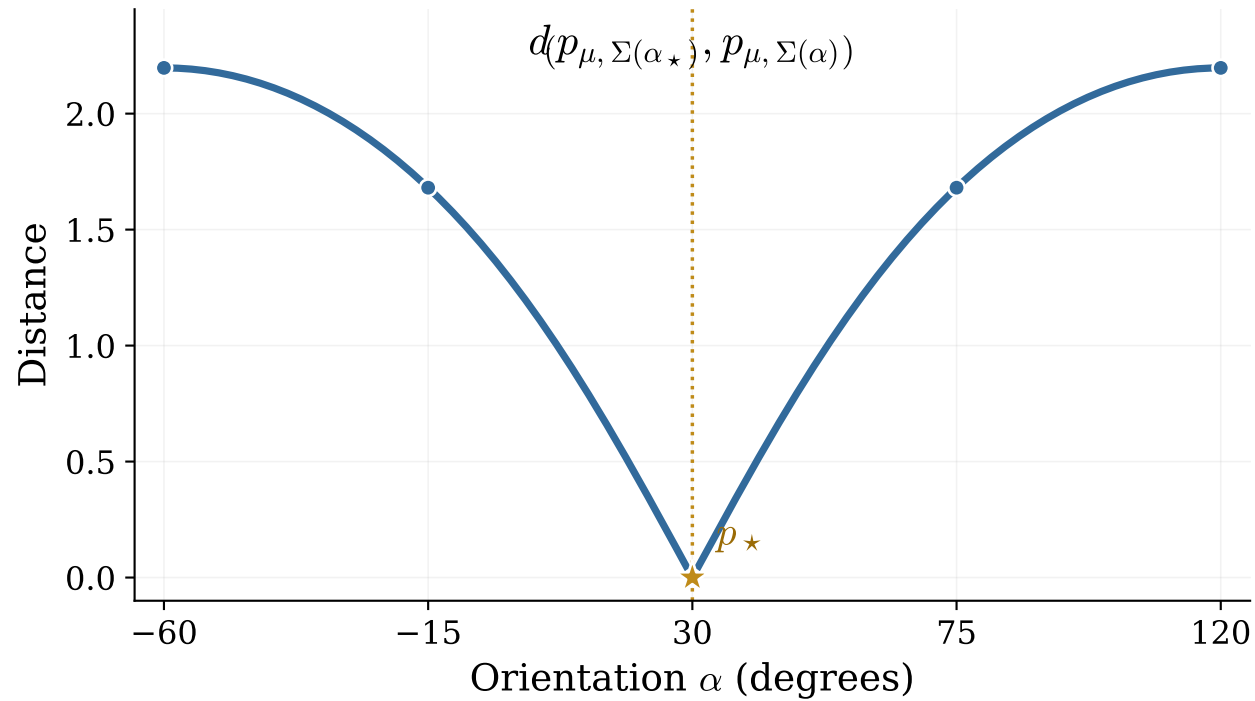


Gaussian distance as the covariance rotates

$$\mu = (0.5, -0.5)^T, \quad (\sigma_1, \sigma_2) = (1.65, 0.55) \quad (\text{fixed}), \quad \alpha_\star = 30^\circ$$

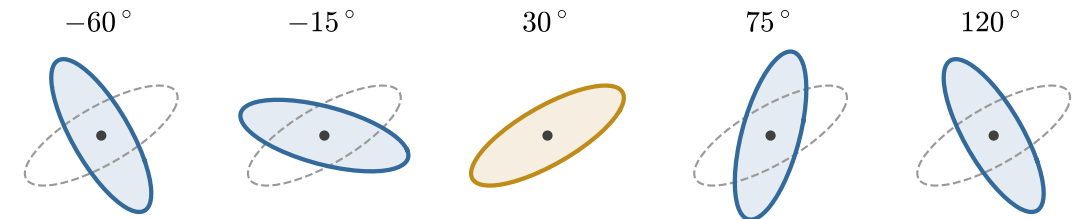
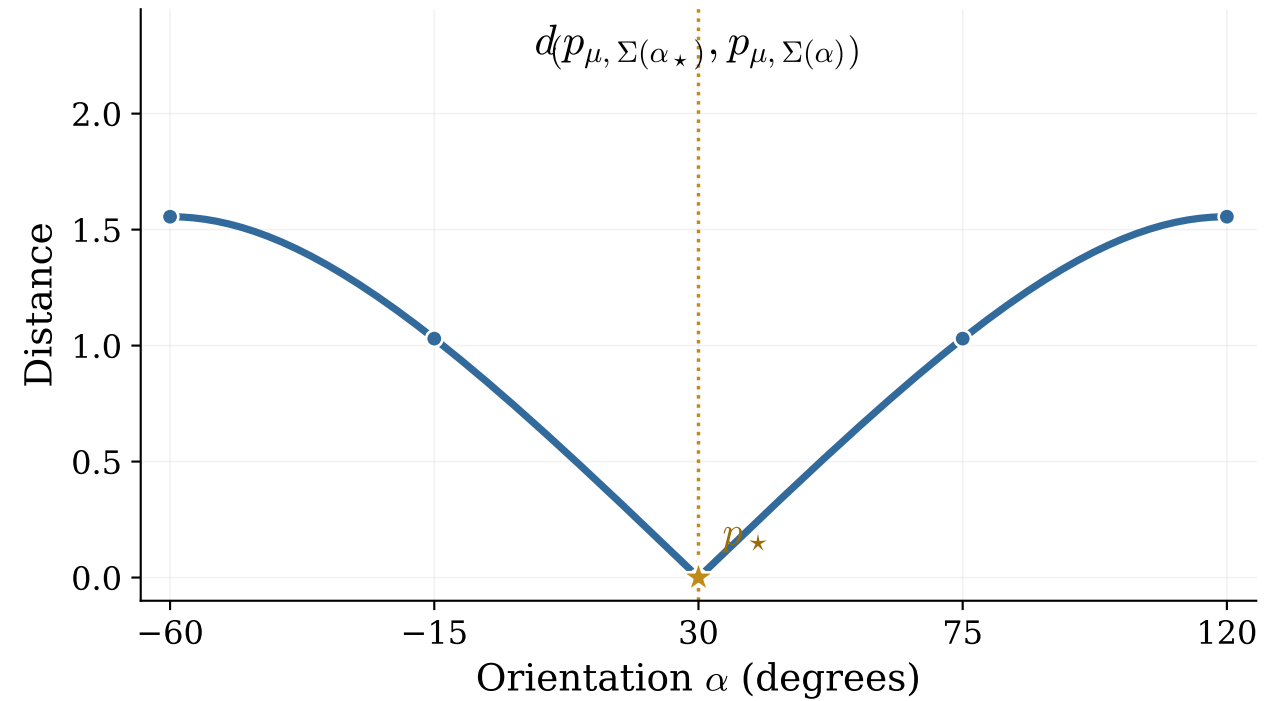
$$\Sigma(\alpha) = R(\alpha) \text{diag}(\sigma_1^2, \sigma_2^2) R(\alpha)^T$$

Fisher-Rao



Density-level ellipses; dashed outline: reference

Wasserstein-Otto



Density-level ellipses; dashed outline: reference

Distances between endpoint Gaussians: the connecting geodesics may change their principal scales.

The distance is symmetric about $\alpha_\star$ and 180° -periodic; the two endpoint orientations describe the same covariance.

Both graphs use identical axes. Ellipses are centered at the common mean and drawn at the same spatial scale.